

Topic 21: Biotechnology and Genetic Modification

21.1 Biotechnology and Genetic Modification

Biotechnology is the use of living organisms or their products for practical purposes. **Genetic modification** involves changing the genetic material of an organism by removing, changing, or inserting individual genes.

Usefulness of Bacteria

Bacteria are particularly useful in biotechnology and genetic modification due to:

- **Rapid reproduction rate:** Bacteria can reproduce quickly, allowing for the large-scale production of desired products.
- **Ability to make complex molecules:** Bacteria can be engineered to produce complex molecules, such as human proteins.
- **Few ethical concerns:** There are generally fewer ethical concerns over the manipulation and growth of bacteria compared to other organisms.
- **Presence of plasmids:** Plasmids are small, circular DNA molecules that can be easily modified and used as vectors to transfer genes.

21.2 Biotechnology

Anaerobic Respiration in Yeast

- **Ethanol for biofuels:** Yeast undergoes anaerobic respiration (fermentation) to produce ethanol, which can be used as a biofuel.
- **Bread-making:** In bread-making, yeast produces carbon dioxide gas, which causes the dough to rise.

Enzymes in Biotechnology

- **Pectinase:** Used in fruit juice production to break down pectin in the fruit cell walls, increasing the juice yield.
- **Lactase:** Used to produce lactose-free milk by breaking down lactose into simpler sugars (glucose and galactose).
- **Biological washing powders:** Contain enzymes (e.g., proteases, lipases) that help break down stains.

Fermenters

Fermenters are large vessels used for the large-scale production of useful products by microorganisms. They provide a controlled environment for the growth of bacteria and fungi.

Controlled conditions in a fermenter:

- **Temperature:** Maintained at the optimal level for the microorganism's enzymes.
- **pH:** Kept at the optimal level for the microorganism's enzymes.
- **Oxygen:** Supplied for aerobic respiration, if required.
- **Nutrient supply:** A continuous supply of nutrients is provided for the microorganisms to grow and reproduce.
- **Waste products:** Waste products are removed to prevent them from inhibiting the growth of the microorganisms.

Products from fermenters:

- **Insulin:** Genetically modified bacteria are used to produce human insulin for treating diabetes.
- **Penicillin:** The fungus *Penicillium* is grown in a fermenter to produce the antibiotic penicillin.
- **Mycoprotein:** A food source produced from the fungus *Fusarium*.

21.3 Genetic Modification

Genetic modification is the process of changing the genetic material of an organism by removing, changing, or inserting individual genes.

Process of Genetic Modification

The process of genetic modification using bacterial production of a human protein as an example involves the following steps:

- 1 **Isolation of the DNA:** The human gene is isolated from human DNA using **restriction enzymes**, which cut the DNA at specific recognition sites, creating **sticky ends**.
- 2 **Cutting of the plasmid:** The same restriction enzymes are used to cut the bacterial plasmid DNA, creating complementary sticky ends.
- 3 **Insertion of the human DNA:** The isolated human gene is inserted into the bacterial plasmid. **DNA ligase** is used to join the sticky ends, forming a **recombinant plasmid**.
- 4 **Insertion of the recombinant plasmid into bacteria:** The recombinant plasmids are inserted into host bacteria.
- 5 **Multiplication of the bacteria:** The bacteria containing the recombinant plasmids are grown in a fermenter, where they multiply rapidly.
- 6 **Expression of the human gene:** The bacteria express the human gene, producing the human protein.